Group II, Row IV: final result, compact form ($\delta^{mm} = d_\perp$, exact limits)

$$\frac{d^3N_{\rm IV}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\int_{x,x',y,y'}\mathcal{O}\Bigg\{\big(d_\perp\bar{E}-2\big)\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}$$
$$\times\left[L\,e^{-ik\cdot(y'-y)}+\int_{\xi_0}^{\xi_1}d\xi\,\frac{\xi^{-2}e^{-ik\cdot(y'-y)/\xi}-e^{-ik\cdot(y'-y)}}{1-\xi}+\int_{\xi_0}^{\xi_1}d\xi\,\frac{1-2\xi}{\xi^3}\,e^{-ik\cdot(y'-y)/\xi}\right]$$
$$+\,d_\perp\int_{\xi_0}^{\xi_1}d\xi\,\frac{1-\xi}{\xi}\,e^{-ik\cdot(y'-y)/\xi}\left[\bar{E}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]\Bigg\},\tag{1}$$

where

$$\bar{E}=\frac{1}{\bar{\epsilon}}-\log\frac{(y'-y)^2\mu^2_{\overline{MS}}e^{2\gamma_E}}{4},\qquad L=\log\frac{\vee}{\Lambda}-\log\frac{\vee}{k^+}+\log\Big(1-\frac{k^+}{\vee}\Big)+\log\Big(1+\frac{\Lambda}{k^+}\Big),$$
$$\xi_0=\frac{k^+}{\vee},\qquad \xi_1=\frac{k^+}{k^++\Lambda},\qquad \mathcal{O}=\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\rho^{e'}(x')\,\Big[U^{ce}(y)-U^{ce}(x)\Big]\rho^{e}(x).$$